

# Simulator to Help Reduce Base Station Operational Cost

Powering base transceiver sites is a challenge that spooks the telecom industry. Here is a simulator that helps operators arrive at the perfect combination of grid power, solar PV, diesel generator and batteries to provide sustainable uninterrupted power to base stations at optimal costs

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One are the times when telecommunications infrastructure was confined to telephone exchanges. With the advent of mobile communications, base stations are almost everywhere. In India, there are over 400,000 sites hosting base stations. All these base transceiver sites (BTS) need to have uninterrupted power supply. Each BTS consumes a few kilowatts of power to run not just the electronics but also the cooling equipment.

Grid power in India is available at a tariff of ₹ 5 to 6 per unit. If this were the predominant source of power, the cost of fuelling a BTS would not be too high. However, India is plagued by power cuts, which could last two to eight hours a day in urban areas and up to 20 hours a day in rural areas. Further, almost 18 per cent of BTS sites in India are off-grid. All this makes backup power systems critical to the functioning of a BTS.

## Bye-bye, band-aid backup

In most cases, power backup at base stations is implemented in an *ad-hoc* fashion, without much scientific planning. Therefore opera-

tional expenditures (OPEX) skyrocket beyond control, and the benefits of renewable energy sources such as photovoltaic (PV) cells are not accrued.

The Indian telecom industry consumes more than two billion litres of diesel and emits over five metric tonnes of carbon dioxide. Energy, power and fuel amount to more than

30 per cent of the total network OPEX. If the telecom infrastructure is to expand in a sustainable fashion, operators will have to turn to an optimal mix of power options that includes renewables also.

Sadly enough, there are not many tools available to help operators configure the power supply for a base station, since developed nations—the so-called fore-runners in the technological evolution—have uninterrupted and ubiquitous grid power. For an indigenous problem, we need an indigenous solution.

A team headed by Padma Shri Dr Ashok Jhunjhunwala at the Reliance-IIT Madras Telecom Centre of Excellence (RITCoE) has risen to the occasion and developed a simulator that helps operators arrive at the optimal mix of power options for their base stations.

## A fresh look at an old problem

RITCoE's report 'Powering Cellular Base Stations: A Quantitative Analysis of Energy Options' is a fresh, new approach, since it does not make any assumptions about the suitability of any source—be it batteries, diesel generator (DG) sets, solar PV cells or grid power. It relies on a

